

Robert G. Gambee

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AI Research Engineer dedicated to improving the world for future generations

Core Competencies

- Proven ability to rapidly master new technologies and adapt in response to evolving needs
- Expertise at driving complex projects from inception to production
- Talent for balancing big-picture strategic thinking and detail-oriented execution
- Experience designing systems to be reliable, scalable and maintainable
- Passion for leveraging technical skills to create meaningful positive impact

Software Skills

Proficient

- Python
- JavaScript, TypeScript, React
- SQL, PostgreSQL, BigQuery
- Git, Jira
- Claude Code, Cursor

Experienced

- PyTorch, scikit-learn
- Docker
- GitHub Actions
- Langfuse, OpenTelemetry
- Bash

Familiar

- Google Cloud Platform
- Amazon Web Services
- Next.js
- Go
- Rust

Professional Experience

FutureSearch: Startup using AI for research and forecasting 2025 to Present
AI Research Engineer

Responsibilities

- Evaluate and enhance capabilities of AI research agents
- Build robust pipelines for running thousands of AI agents concurrently
- Work throughout the stack to rapidly iterate on product ideas
- Log data and build dashboards to monitor performance, stability and cost
- Keep up to date with latest developments in AI capabilities and tools
- Communicate with users to get their feedback and help them get the most out of our product

Technical Projects

- Tech Lead for tools to let AI research agents search the internet and read documents 2025 to Present
 - Developed and evaluated tools to answer queries about web pages
 - Extensively analyzed agent transcripts to identify failure modes
 - Eliminated leading failure mode by switching to superior page retrieval service
 - Optimized page reading flow to reduce maximum latency by factor of 5
- Web app for orchestrating thousands of AI agents to research hard questions at scale 2025 to Present
 - Owned features for importing, viewing and exporting tabular data
 - Designed and built UI for inspecting every step an agent took to produce its answer
 - Prioritized transparency and visibility to help users understand and trust the results
- Open-source library for interfacing with our services 2026 to Present
 - Designed API, SDK and MCP server to be easy to use yet powerful
 - Managed process for publishing releases
- AI validator for uncertain claims 2025
 - Developed workflow to search the internet for relevant information
 - Created judge to weigh evidence and estimate the probability that the claim is true
 - Wrote and maintained a suite of challenging evaluations
 - Benchmarked custom validator against other approaches across many models

MIT FutureTech – AI Risk Initiative

2026 to Present

Software Engineer (Part-Time Contractor)

Technical Projects

- Python library for building document processing pipelines 2026 to Present
 - Created suite of reusable components to accelerate building of future pipelines
 - Wrote extensive unit tests, automated checks and AI guidelines to promote correctness and consistency
- Automated pipeline for collecting and classifying AI risks 2026 to Present
 - Designed and built pipeline to process over 10,000 documents, vastly more than previous manual one
 - Crafted LLM prompts to screen documents, balancing false positives against false negatives
 - Engineered retry and resume features to make pipeline robust to errors
 - Optimized pipeline to balance accuracy, cost and speed

Formlabs: Industry leader in professional 3D printing

2015 to 2025

Systems Integration Engineer III

Responsibilities

- Architected software which was maintainable, scalable and testable
- Owned key printer systems for the entire product cycle, driving them from inception to public release
- Understood complex interactions between printer systems, keeping both details and big picture in mind
- Optimized for printer reliability through robust design and failure mode prediction
- Analyzed and visualized printer data to answer pressing questions and inform business decisions
- Mentored junior team members to foster their technical abilities

Technical Projects

- Senior embedded developer for Form 4 and Form 4L 2021 to 2025
 - Thoughtfully crafted powerful yet understandable API to control all aspects of product functionality
 - Comprehensively audited API for security vulnerabilities and coordinated plan to address them
 - Advised architectural decisions for embedded and desktop software
- Data management on user-replaceable components for Form 4 and Form 4L 2023 to 2024
 - Designed a unified architecture for all components, agnostic to interface and data format
 - Wrote extensive validation checks to be robust to failures when reading or writing
 - Secured system against counterfeiting to protect company's primary revenue stream
 - Thoroughly tested all code with automated checks
- Print preparation routine for Form 4 and Form 4L 2021 to 2024
 - Sped up routine by a factor of 5 to 10 compared to previous product, vastly improving user experience
 - Wrote predictive checks to give user advance warning of issues and avoid interrupting prints
 - Implemented specification for how to handle over 50 possible errors
- Prototype firmware for early iterations of Form 4 2021 to 2022
 - Independently developed prototype firmware in Python to support crucial conceptual testing
 - Balanced competing desires for flexibility and stability using a modular design
 - Rapidly responded to feature requests and bug reports, addressing them in days if not hours
- Dashboard for plotting live sensor data 2021
 - Independently developed over four days during company hackathon
 - Wrote backend in Go, wrote frontend in JavaScript, streamed data via WebSockets
 - Recognized by the CEO in a company-wide email as one of the most impressive projects that year

Achievements

- Recipient of Formlabs' Perform Award, which recognizes top 10% of employees 2020 and 2023

Independent AI Research

Do LLMs have stable preferences?

2025 to 2026

- Built upon findings in Claude 4 System card to understand how preferences vary under different conditions
- Used pairwise comparisons between tasks to determine the LLM's implicit rating for each task
- Compared multiple models, response formats, orders and numbers of tasks
- Found that ratings for individual tasks combine coherently when considering sequences of tasks
- Uncovered strong dependency on response format: free-form vs. structured output
- Results have implications for AI welfare, highlighting challenges with satisfying LLM preferences

Reproduction of “The Capacity for Moral Self-Correction in Large Language Models”

2023

- Loaded and processed tens of thousands of samples from three different datasets
- Submitted API requests asynchronously, with automatic retries and rate limiting
- Analyzed bias in model responses according to three different metrics
- Compared and contrasted results to demonstrate influence of RLHF training vs. prompt engineering

Volunteer Experience

AI Governance and Safety Canada

2024

- Designed and implemented flexible system for scraping information relevant to the AI safety community
- Leveraged LLM to robustly scrape many sites with zero site-specific tuning required
- Initially developed to find upcoming events, but can be extended to other types of content, e.g. publications
- Created automated workflow using GitHub Actions to run scraper and publish output to database
- Overhauled collection of introductory AI resources with updated list covering many topics and formats

Personal Projects

Chronicle

2023

Web app to keep track of how one spends one's time

- Used Django framework to manage HTTP requests and access SQLite database
- Presented data as a table for sorting and filtering, as well multiple charts for visualization
- Set up automated test and deployment workflows using GitHub Actions

SCAFFOLD

2023

Completed as part of AI Safety Camp (3 person team plus advisor)

- Built React web app to generate feedback on one's research ideas using GPT
- Fine tuned model to make its responses more relevant to AI safety research

Publications

Deep Research Bench: Evaluating AI Web Research Agents (2025). FutureSearch: Nikos I. Bosse, Jon Evans, **Robert G. Gambee**, Daniel Hnyk, Peter Mühlbacher, Lawrence Phillips, Dan Schwarz, Jack Wildman. <https://doi.org/10.48550/arXiv.2506.06287>

Barium titanate nanoparticles: Short-range lattice distortions with long-range cubic order (2018). Chenyang Shi, Simon J. L. Billinge, Eric Puma, Sun Hwi Bang, Nathaniel J. H. Bean, Jean-Claude de Sugny, **Robert G. Gambee**, Richard C. Haskell, Adrian Hightower, Todd C. Monson. <https://doi.org/10.1103/PhysRevB.98.085421>

Permittivity effects of particle agglomeration in ferroelectric ceramic-epoxy composites using finite element modeling (2018). Jonas L. Kaufman, Scott H. Tan, Kirkclann Lau, Ashka Shah, **Robert G. Gambee**, Chris Gage, Lupe MacIntosh, Albert Dato, Peter N. Saeta, Richard C. Haskell, Todd C. Monson. <https://doi.org/10.1063/1.5053442>

Education

Harvey Mudd College, Claremont, CA

Bachelor of Science in Engineering with High Distinction

2011 to 2015

- GPA: 3.8
- Inducted into Tau Beta Pi, national engineering honor society
- Recognized on Dean's List of top performing students

2014

2012 to 2015

Undergraduate Projects

SpaceX, Hawthorne, CA & Harvey Mudd College

2014 to 2015

Recoverable Flight Data Recorder (5 person team)

- Designed housing and selected materials to protect electronics from rocket explosion
- Built and tested prototypes according to SMC-S-016 and other specifications
- Contributed to software for receiving flight data over UDP and saving to SD card

Academic Research, Harvey Mudd College

2014 to 2015

Gas Permeation Across Nanocomposite Polymer Membranes (5 to 8 person team)

- Performed gas permeation experiments on synthesized membranes
- Ran and analyzed molecular simulations containing over ten thousand atoms each
- Wrote grant proposal for Amazon EC2 resources that decreased runtime by an order of magnitude

Sandia National Laboratories, Albuquerque, NM & Harvey Mudd College

2013 to 2014

Measurement of Barium Titanate Nanoparticle Permittivity (5 person team)

- Developed analytical and numerical models for interpretation of experimental data
- Presented work at Materials Research Society conference as an invited speaker
- Project findings were later published in American Institute of Physics Advances